

# ANISH SIDDIQUI

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## SUMMARY

Innovative Computer Engineering and Analytics graduate with experience in digital systems, hardware development, and electromechanical integration, seeking full-time employment.

## EDUCATION

**University of Houston – Cullen College of Engineering | Houston, TX**

**Graduation Date:** May 2025

**Bachelor of Science in Computer Engineering and Analytics**

**GPA:** 3.549

**Relevant Coursework:** *Pattern Recognition & Machine Learning, Microprocessor Systems, Embedded Computing & IoT*

## EXPERIENCE

**Enterprise Products – Information Technology Intern**

May 2024 - August 2024

- Partnered with department supervisors to analyze asset functionality and construct profit outlook portfolio.
- Accelerated technical operations by designing and deploying specialized applications in eBusiness system solutions.
- Built transaction control language queries in MySQL Enterprise to extract information from relational databases.
- Used cloud infrastructure services to test program capability, integrity, and compatibility in Oracle APEX.

**NRG Energy – Data Science Intern**

May 2023 - August 2023

- Implemented batch processing scripts in Jupyter Notebook to frame predictive analytics dashboards.
- Constructed neural networks using NumPy and Pandas coding packages to visualize clientele decision probability
- Designed and presented detailed evaluation capstone to demonstrate AI extrapolation capabilities.
- Improved stream ingestion architecture to refine non-linear continuous databases leading to qualitative discrete categorizing

**CVS Pharmacy – Pharmacy Technician**

January 2023 - May 2023

- Collaborated with licensed Pharmacist to ensure proper preparation, storage, and purchase of designated medications.
- Administered over one hundred Covid-19 tests and scheduled appropriate vaccinations for eligible patients.

## PROJECTS

**Custom FPGA Embedded Systems Platform**

May 2026

- Developed an FPGA-based logic analyzer using Verilog HDL to capture and analyze real-time digital signals.
- Implemented and verified RTL modules for signal sampling, triggering, and SPI protocol decoding.
- Modeled a custom carrier PCB in KiCAD with GPIO, UART, and sensor interfaces for rapid embedded systems prototyping.
- Utilized Linux shell and Tcl scripts to automate FPGA synthesis, verification, and implementation workflows.

**3D Printed Autonomous Mobile Rover**

December 2025

- Designed a rover chassis in FreeCAD inspired by NASA Lunar Terrain Vehicle (LTV) project requirements.
- Integrated DC motors and servo actuators with IMU feedback and regulated power electronics to allow autonomous driving.
- Enabled wireless user control and live onboard camera streaming using multiple microcontroller systems.
- Fabricated a rotating LiDAR-guided rover platform built to support SLAM-based mapping and localization.

**Dynamic Multimodal Robotic Arm Control System**

June 2025

- Assembled OWI-535 Robotic Arm Edge connected to specialized PCB allowing users full 6DOF manual control.
- Configured MATLAB vision system with Logitech Pro webcam for autonomous object tracking using color detection.
- Deployed modified Vosk speech engine alongside Python and Arduino for offline voice-based control.

**Customized H-Bridge Motor Driver**

April 2025

- Engineered discrete MOSFET-based H-bridge motor driver for bidirectional DC motor control.
- Structured a four-transistor topology to enable forward, reverse, and braking operation through controlled current flow.
- Applied CMOS design methodologies to analyze transistor switching, gate-drive behavior, and power efficiency.
- Programmed embedded control firmware to generate PWM signals and regulate system operations.

## SKILLS

- **CAD/EDA:** *Altium, KiCad, EasyEDA, FreeCAD, LTspice, Onshape, SolidWorks*
- **Programming:** *C++, Verilog HDL, Tcl, Shell Scripting, SQL, PL/SQL, MATLAB, Java, ARM Assembly*
- **Platforms & Tools:** *Linux, Oracle APEX, Databricks, JupyterLab, ROS 2, Power BI*
- **Hardware:** *STM32 Microcontrollers, ESP32/ESP8266, TI LaunchPad, Arduino, Elegoo Uno R3*

## CERTIFICATIONS

**Global Electronics Association:** *IPC/WHMA A-620 (Cable and Wire Harness Assemblies)*

April 2026

**Oracle University:** *Oracle Cloud Infrastructure AI Foundation Associate*

February 2026

**Global Electronics Association:** *IPC J-STD-001 (Soldered Electrical and Electronic Assemblies)*

January 2026